TRACKIFY:

DIGITAL PLATFORM FOR STUDENT ASSIGNMENT AND PROJECT MANAGEMENT

Phase 3 - Analysis and Design

Group Mario

Section 03

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SECD2613 System Analysis and Design

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1.0 Overview of The Project

Phase 3 of the Trackify project focuses on the design and modelling of the proposed system. Based on the findings from Phase 2, where issues such as missed deadlines, poor communication, and lack of progress tracking were identified, this phase presents the system design and demonstrates how Trackify can help improve project coordination and monitoring.

The objective of this phase is to transform the requirement gathered earlier into a clear system design. This includes defining the TO-BE workflow, data flow diagrams, CRUD matrix and event-response tables. These elements illustrate how Trackify will operate as a centralized platform, enabling students to register accounts, manage assignment s, receive smart reminders, track progress visually and collaborate effectively in groups. Administrators will maintain oversight by managing accounts, assignments and group records to ensure accountability and data integrity.

By presenting both the current (AS–IS) and future (TO–BE) workflow, Phase 3 demonstrate how Trackify evolves from concept into a practical solution. The system design highlights usability, security and accessibility ensuring that Trackify is not only functional but also for long-term academic use.

2.0 Problem Statement

2.1 Busy Schedules and Time Management Issues

Packed timetables and involvement in university programs or associations are challenging for students to allocate time effectively.

2.2 Communication Gaps

For group projects, there may be lack of communication between the team members regarding project progress for divided tasks.

2.3 Missed Deadlines

Students may forget assignment deadlines or struggle with tracking multiple assignments from different classes.

2.4 Difficulty with E-Learning Systems

The system only shows the deadlines when students are logged in which makes it difficult and inconvenience in some cases. It also does not provide a progress tracker for progress monitoring.

2.5 Lack of Assignments Progress Monitoring

Students lack proper tools to visualize progress and track task completion.

3.0 Proposed Solutions

3.1 Smart Reminders

Popped up notification on the user's smartphone or tablet. Student does not have to login into e-learning and find the assignment deadlines for each subject manually.

3.2 Assignment Progress Tracking

Displaying the percentages of work done with a to do list.

3.3 Group Assignment Tracking

Team members will be able to communicate with each other more efficiently and see their real time progress.

3.4 Priority Ranking System Mode

Assignment or project with the highest priority, whether it be the closest due date and such, will automatically move to the top of the assignment list.

4.0 Current Business Process/Workflow

4.1 Current Business Process

Based on the survey conducted, students currently use several methods to keep track of assignment deadlines, including:

- Lecturer or friends
- E-learning platforms
- Personal notes
- Calendar reminders
- Email notification from graduate.utm.my email

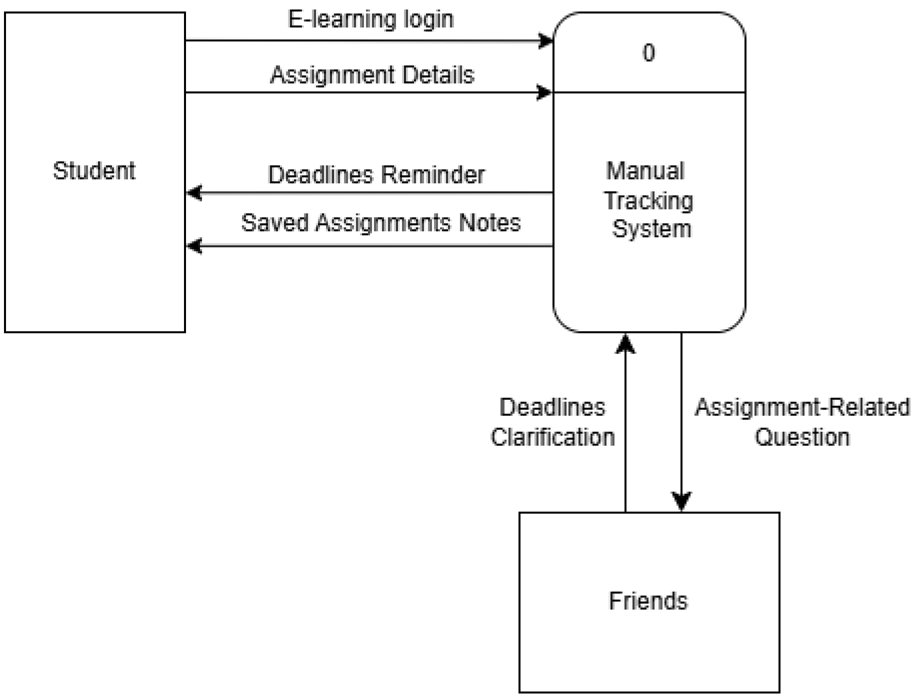
Lecturer will usually inform the students when the submission for assignment is open in e-learning with the deadline, while e-learning will give students reminder on the dashboard section when the due is less than a week. The calendar on the e-learning marks some of the important assignments date but sometimes it can miss some assignment. An email notification from student's graduate email will pop up a few days before the deadline and after the student submitted their work. Some students also write their assignments deadlines on their notebook or smartphone. For group assignments or project, student discusses task in group chat (usually via Telegram or WhatsApp) and the tasks are distributed through messages.

Workflow:

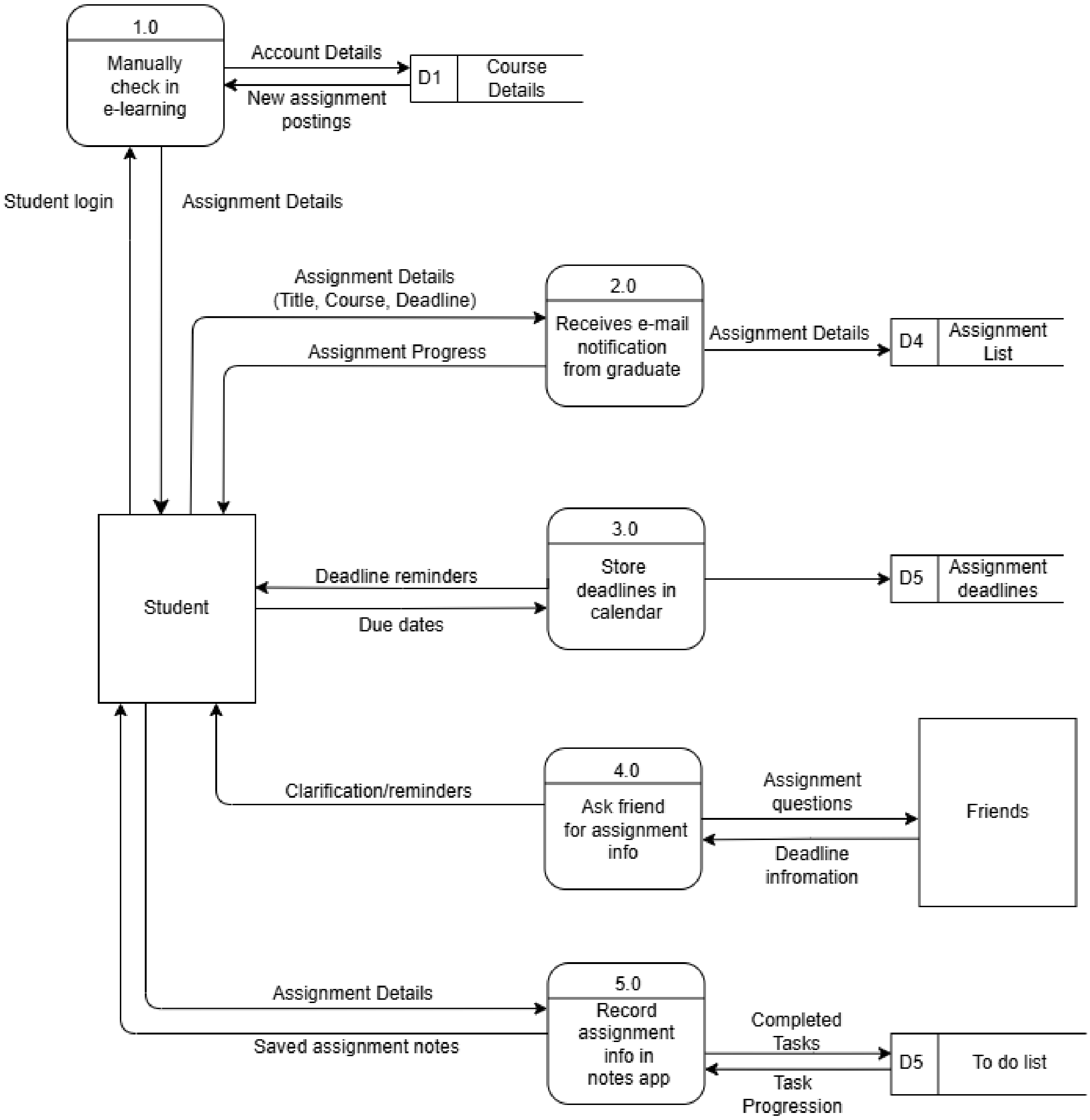
1. Lecturer announces assignment and deadline.
2. Student checks the assignment details on e-learning or saves in their personal notes.
3. Email notification pops up few days before the deadline.
4. For group work, student discusses task in group chat.
5. Tasks are distributed through messages.
6. Progress updates are shared informally in chat.
7. Students ask and remind each other about deadlines.

5.0 Logical DFD (AS-IS)

5.1 Context Diagram



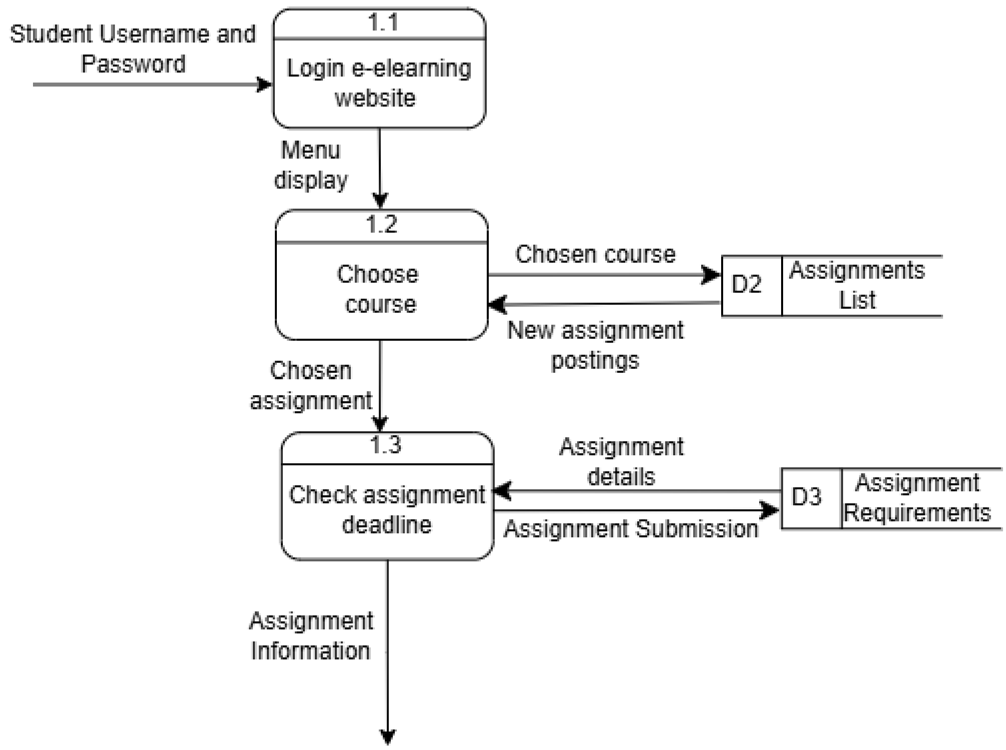
5.2 Diagram 0



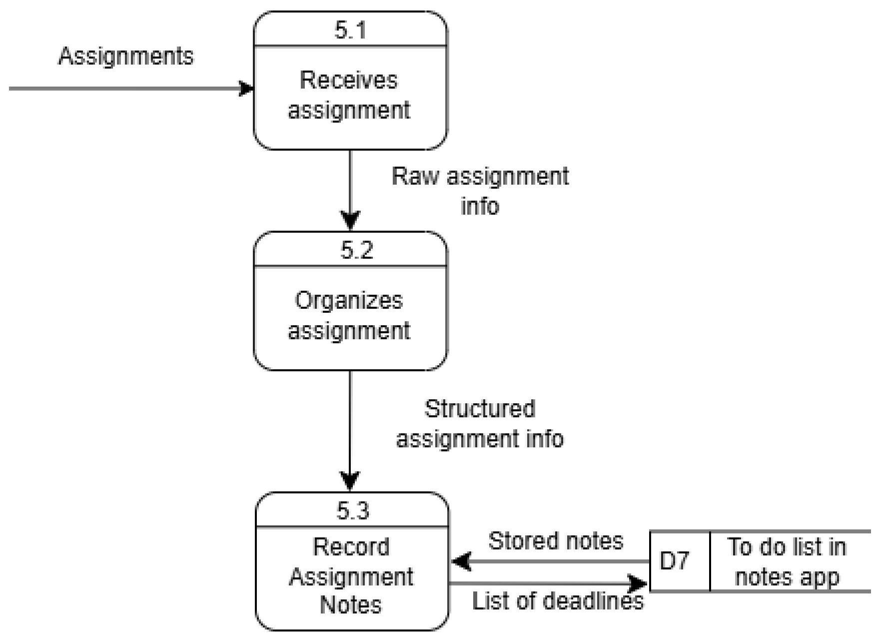
1.0
Manually check in e-learning
Account Details
New assignment postings
D1
Course Details
Student login
Assignment Details
Assignment Details (Title, Course, Deadline)
Assignment Progress
2.0
Receives e-mail notification from graduate
Assignment Details
D4
Assignment List
Student
Deadline reminders
Due dates
3.0
Store deadlines in calendar
D5
Assignment deadlines
Clarification/reminders
4.0
Ask friend for assignment info
Assignment questions
Deadline infromation
Friends
Assignment Details
Saved assignment notes
5.0
Record assignment info in notes app
Completed Tasks
Task Progression
D5
To do list

5.3 Child Diagram

5.3.1 Child Diagram Process 2.0: Manually checks in e-learning



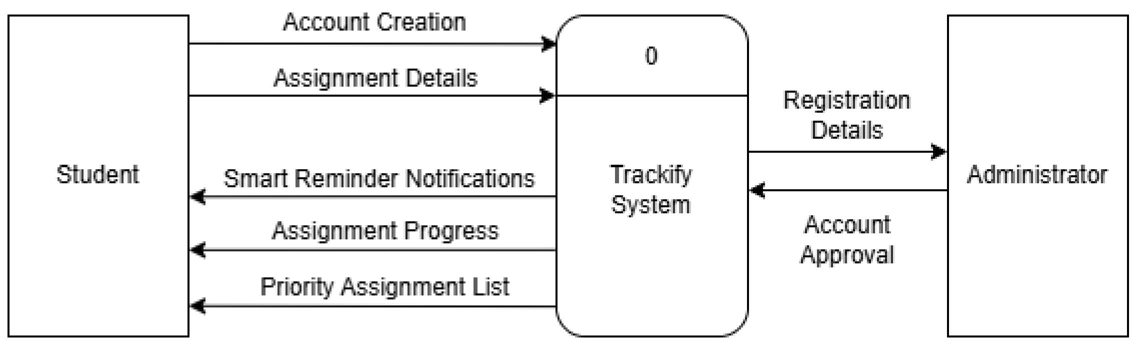
5.3.2 Child Diagram Process 5.0: Record Assignment info in Notes app



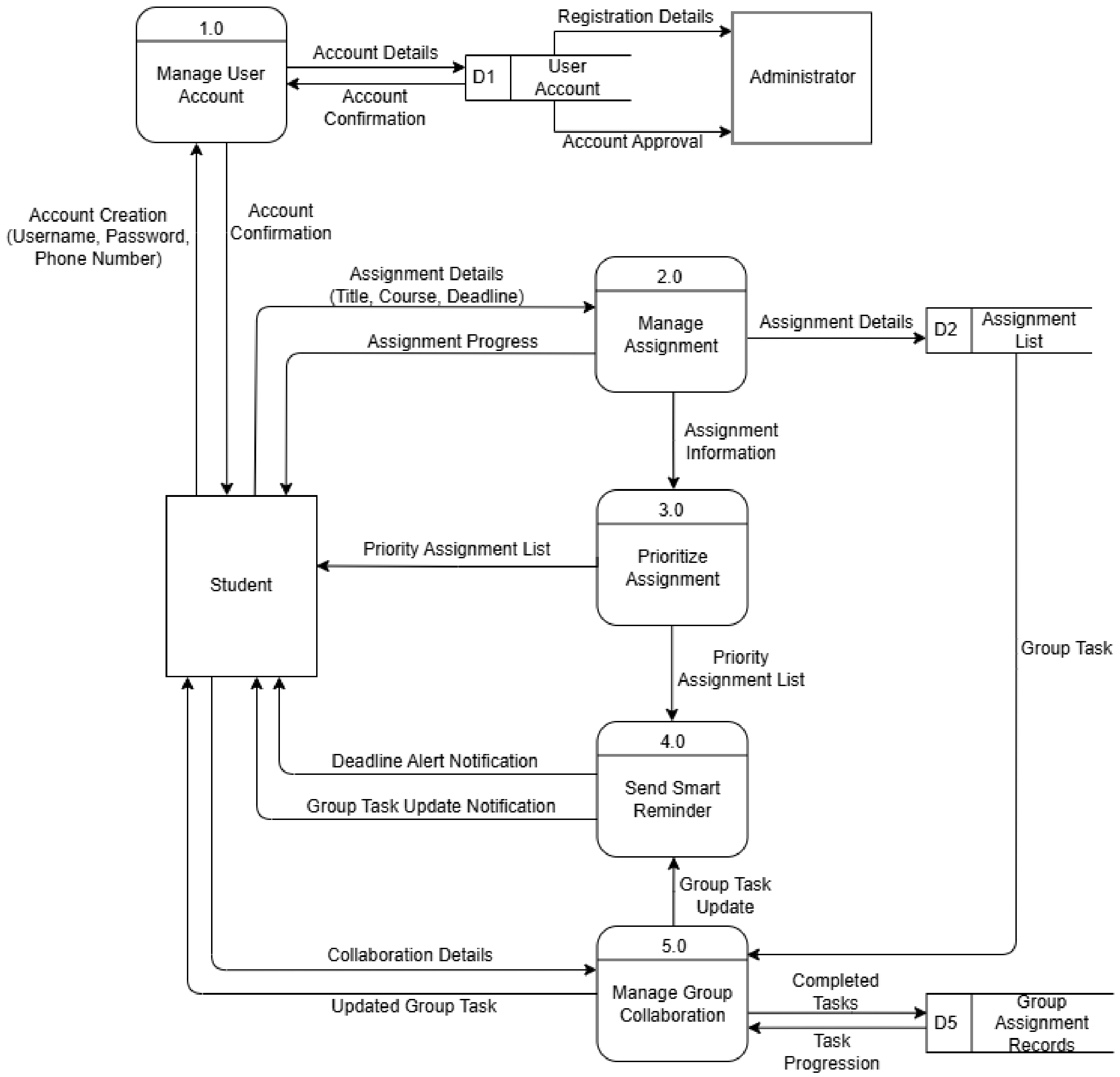
6.0 System Analysis and Specification

6.1 Logical DFD TO-BE system

6.1.1 Context Diagram

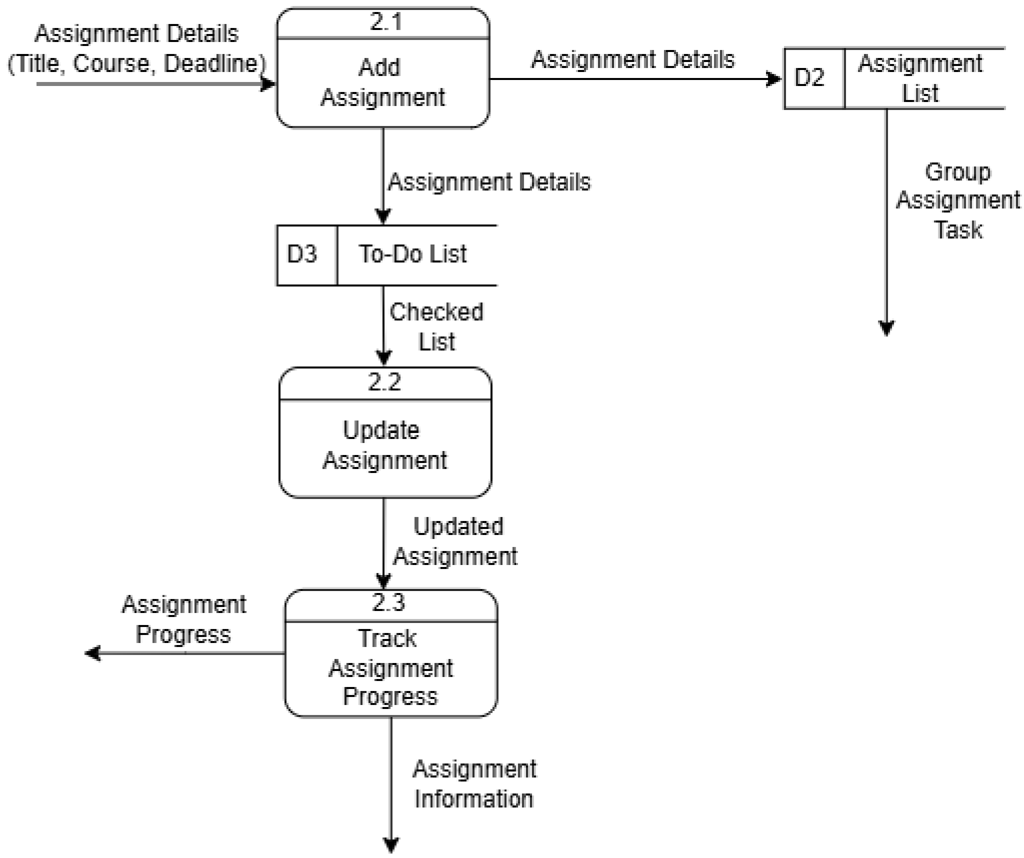


6.1.2 Diagram 0

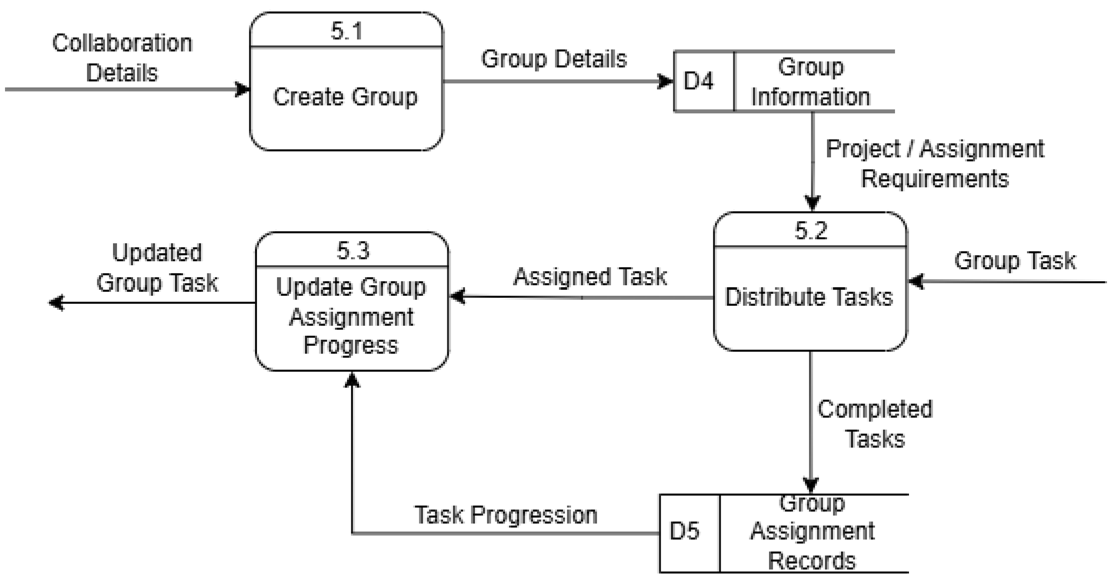


6.1.3 Child Diagram

6.1.3.1 Child Diagram Process 2.0: Manage Assignment



6.1.3.2 Child Diagram Process 5.0: Manage Group Collaboration



6.2 Process Specification

6.2.1 Structure English (2.0 Manage Assignment)

INPUT Assignment Details (Title, Course, Deadline)

IF assignment details are complete THEN

    Store assignment details in Assignment list

    Display assignment information

ELSE

    Display error message

ENDIF

IF user updates assignment details THEN

    Update assignment record

    Store updated assignment details in To-Do list

ENDIF

IF assignment progress is updated THEN

    Record assignment progress

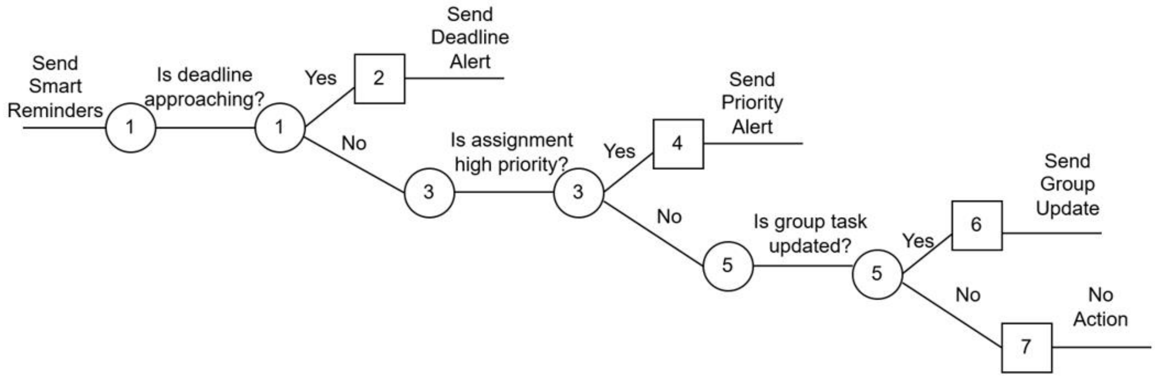
    Display current progress status

ENDIF

6.2.2 Decision Table (5.0 Manage Group Collaboration)

Conditions	**Rules**			
	1	**2**	**3**	**4**
Group Exists?	N	Y	Y	Y
Task Assigned?	N	N	Y	Y
Task Completed?	N	N	N	Y
Actions				
Create Group	X			
Distribute Task		X		
Update Task Progress			X	
Store Completed Task			X	X

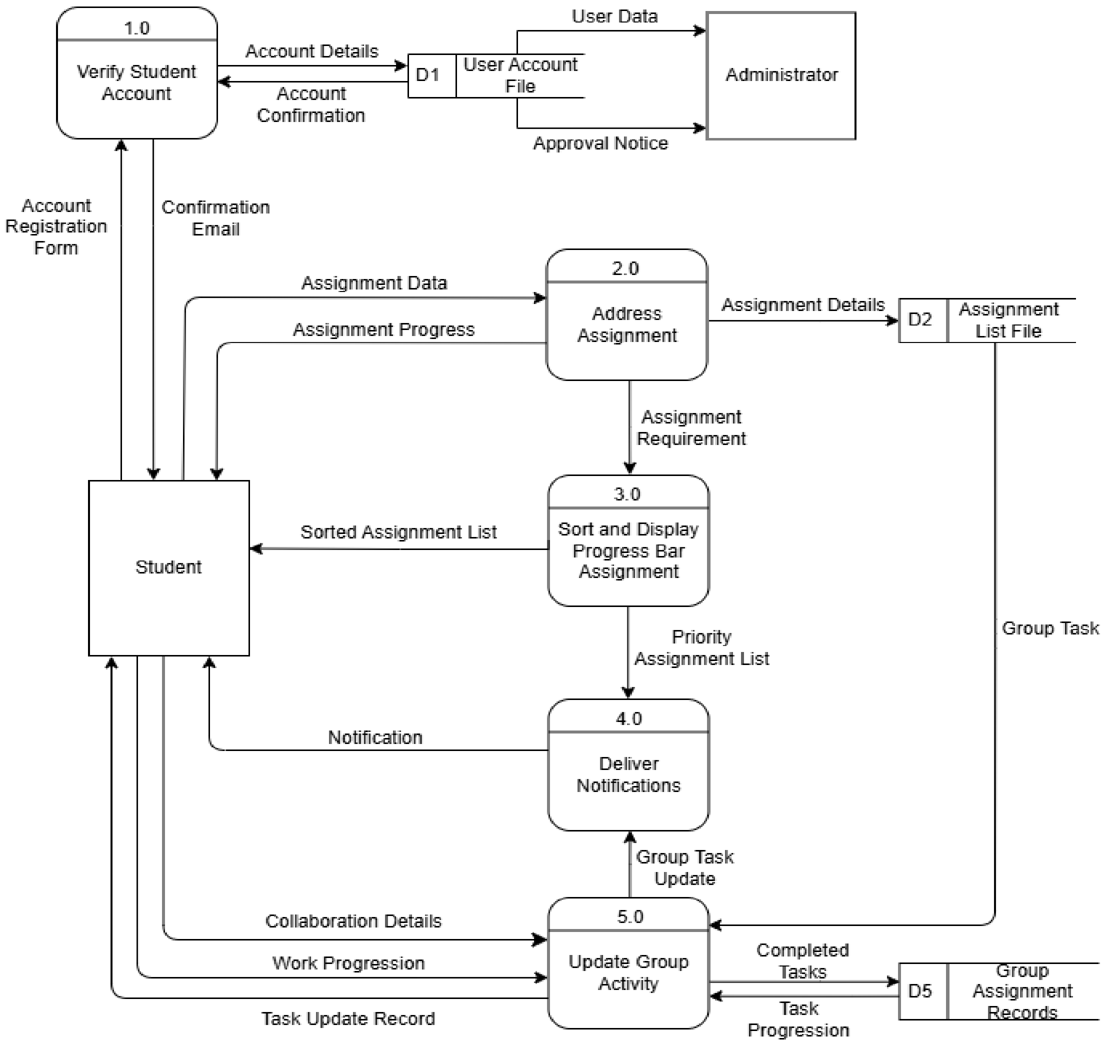
6.2.3 Decision Tree (4.0 Send Smart Reminders)



7.0 Physical System Design

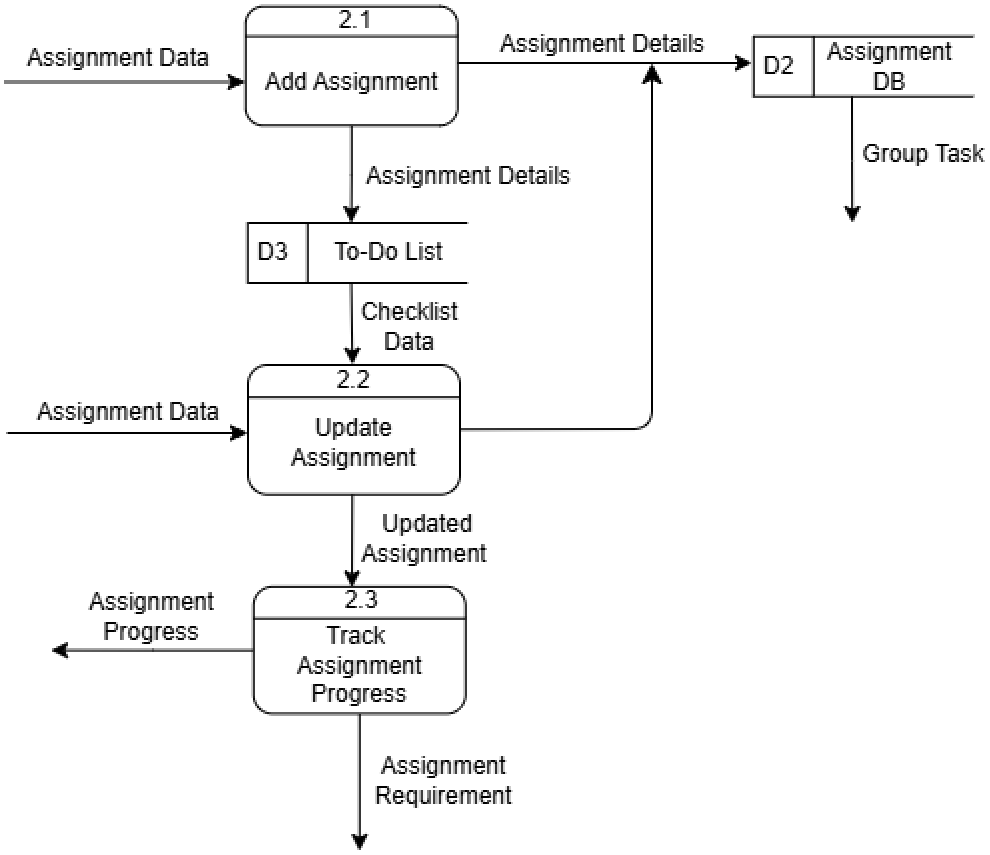
7.1 Physical DFD TO-BE system

7.1.1 Diagram 0

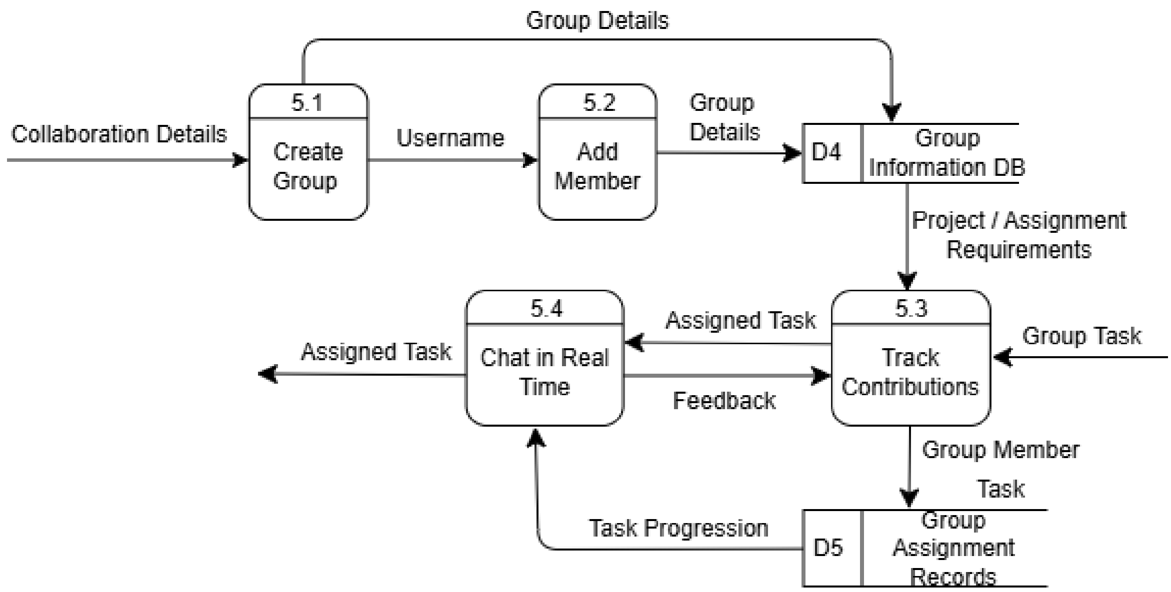


7.1.2 Child Diagram

7.1.2.1 Child Diagram Process 2.0: Address Assignment

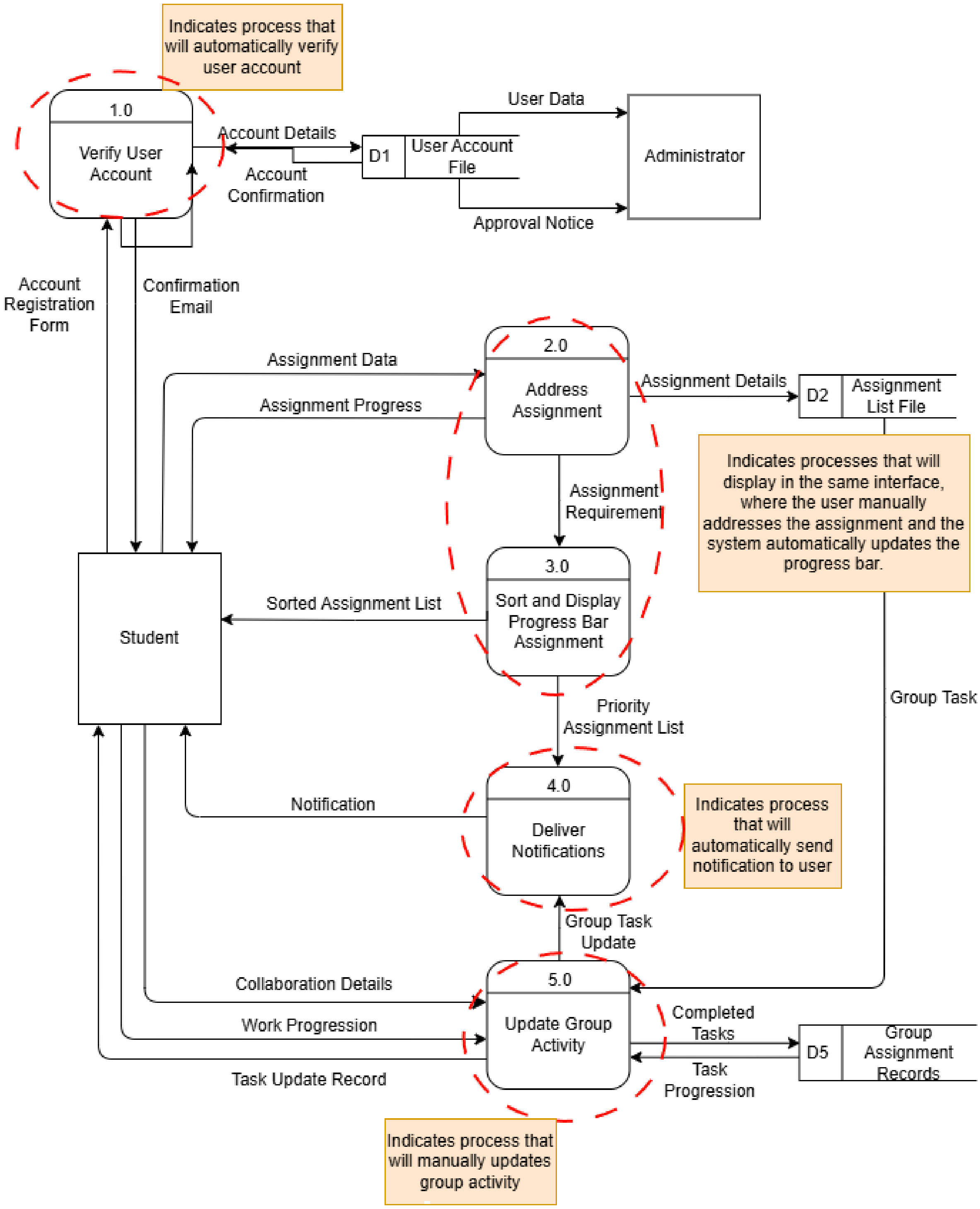


7.1.2.3 Child Diagram Process 5.0: Update Group Activity



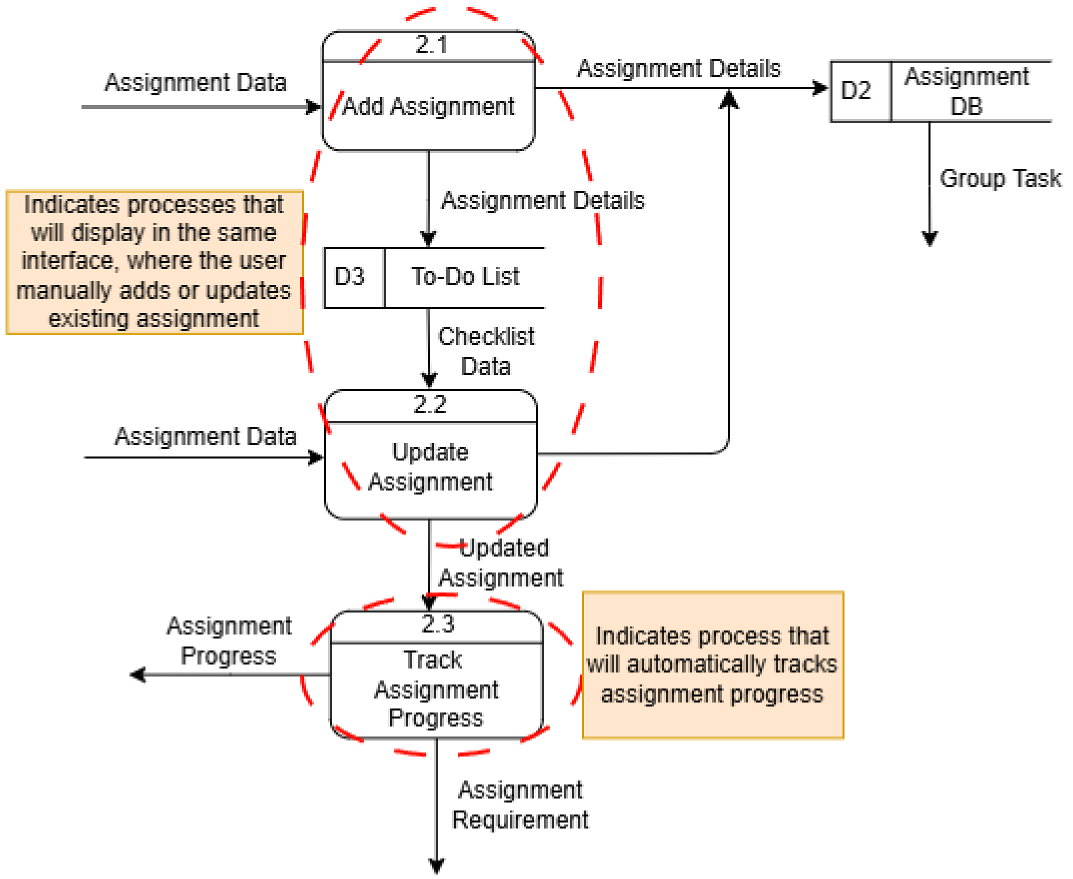
7.2 Partitioning

7.2.1 Diagram 0

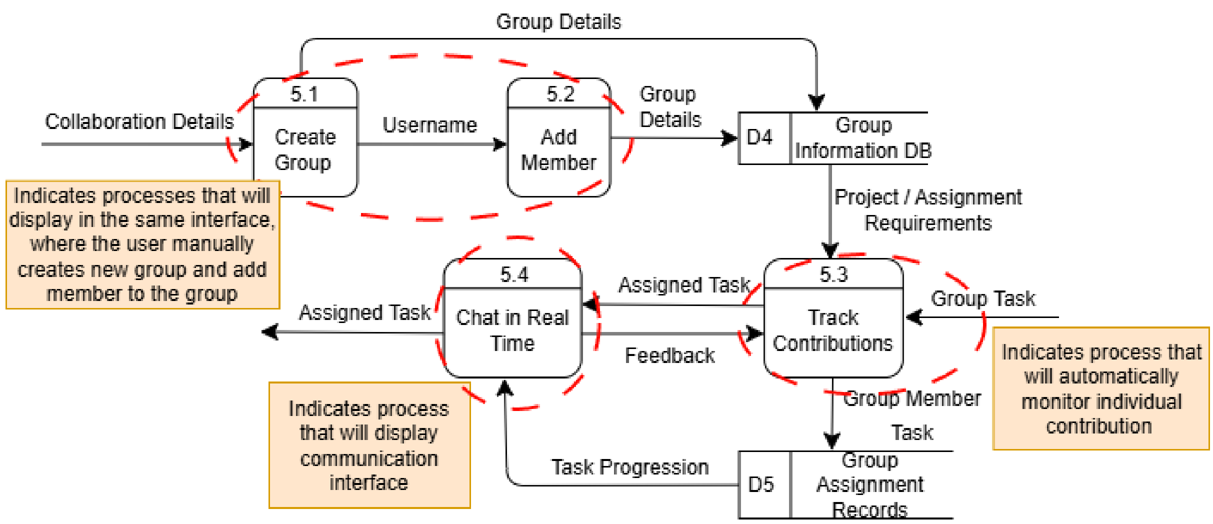


7.2.2 Child Diagram

7.2.2.1 Child Diagram Process 2.0: Address Assignment



7.1.2.3 Child Diagram Process 5.0: Update Group Activity



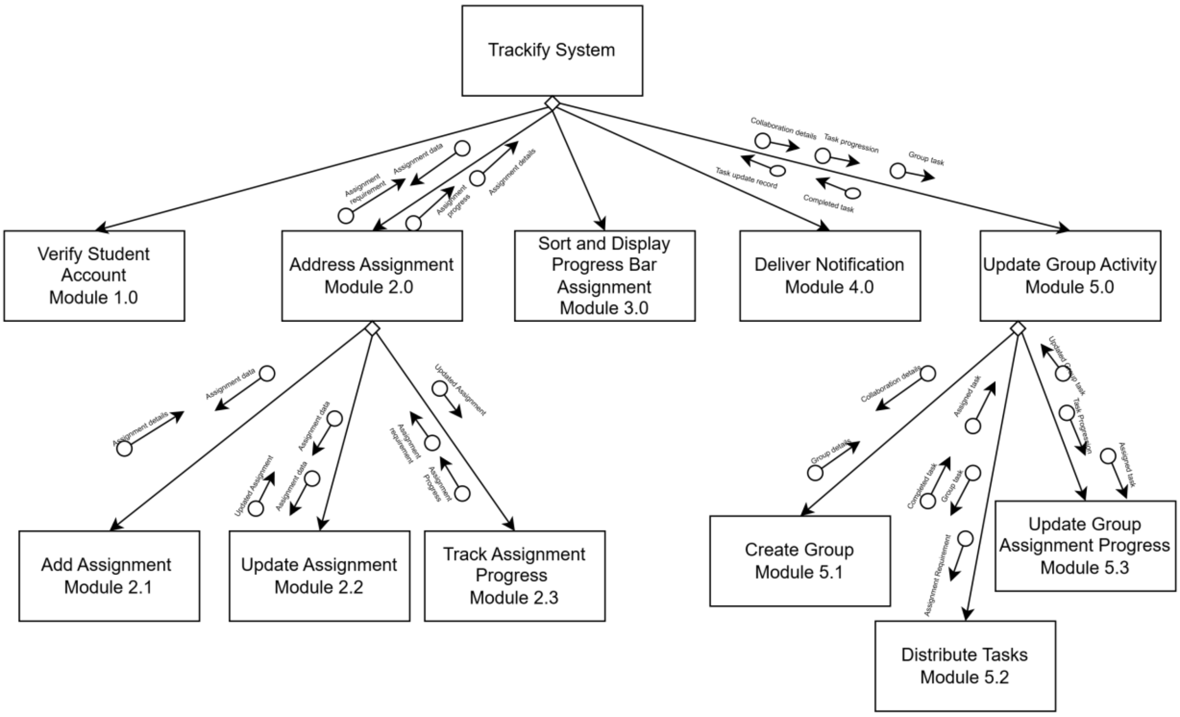
7.3 CRUD Matrix

Entity	Student	Administer
Register/Login Account	R, U	C, R, U, D
Manage Assignment	C, R, U	C, R, U, D
Group Collaboration	C, R, U	R, D
Smart Reminder Notification	R, U	C, R, U, D
Priority Ranking	R	U
Assignment Progress	U	R

7.4 Event Response Table

Event	Source	Trigger	Activity	System Response	Destination
Student register account	Student	Enter username, password, phone number	Register Account	System creates new, user record, confirm registration	Student
Students log in	Student	Submit login credentials	Login Authentication	System verifies, grant access	Student
Students create assignment	Student	Enter assignment details	Manage assignment	System stores assignment record	Assignment Database
Student update assignment process	Student	Update completion percentage	Update Progress	System updates progress bar and save percentage	Student dashboard
Deadline approaching	System	Time-based trigger	Smart Reminder Notification	System sends deadline alert	Student device
Group leader creates group task	Student	Create group and assign tasks	Group Collaboration	System record group task and distributed to member	Group Database
Administration deletes assignment / group	Admin	Select record to delete	Manage Assignment/Group	System removes record from database	Database
Student views assignment list	Student	Open assignment dashboard	Priority Ranking	System display assignment sorted by due date	Student
Group member share update / comment	Student	Post update in group	Group Collaboration	System logs contribution and updates group progress	Group Dashboard

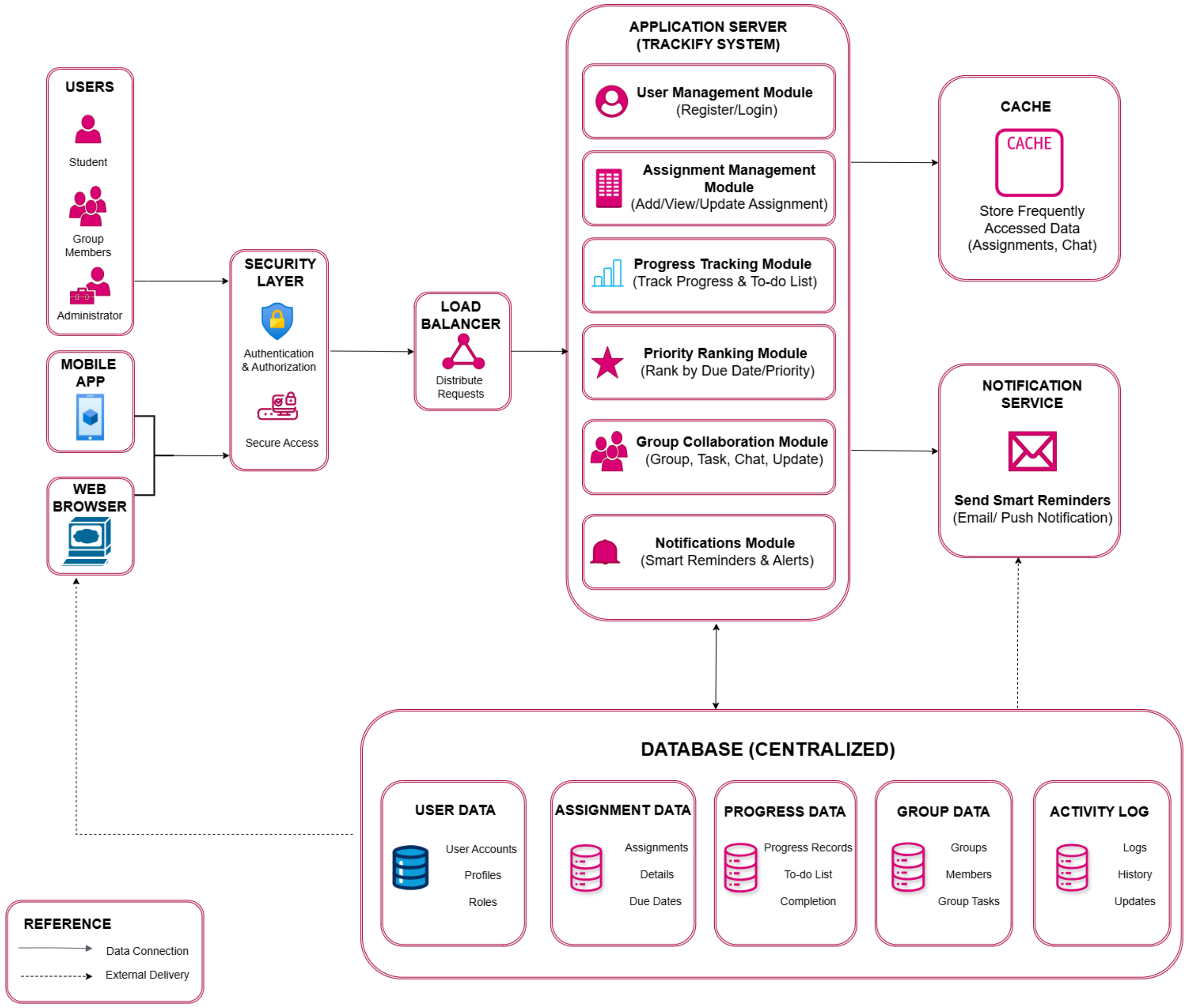
7.5 Structure Chart



7.6 System Architecture

The Trackify system architecture is based on a layered client-server architecture. Users, including students, group members, and administrators, access the system through a web or mobile interface. Requests are first processed through a security layer for authentication and user verification, then forwarded to a load balancer that distributes traffic across application servers.

The application server handles core functions such as user management, assignment management, progress tracking, priority ranking, group collaboration, and notifications. A caching layer improves performance by storing frequently accessed data, while a centralized database stores all system data, including users, assignments, progress records, and group information. This architecture ensures Trackify is secure, efficient, and scalable.



8.0 System Wireframe

8.1 Input Design

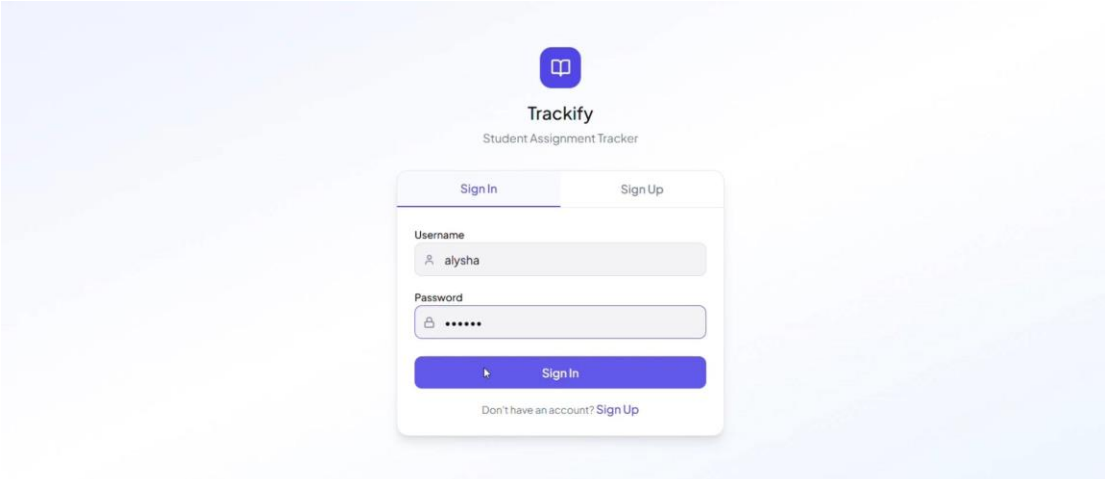


Figure 1 : Enter username and password

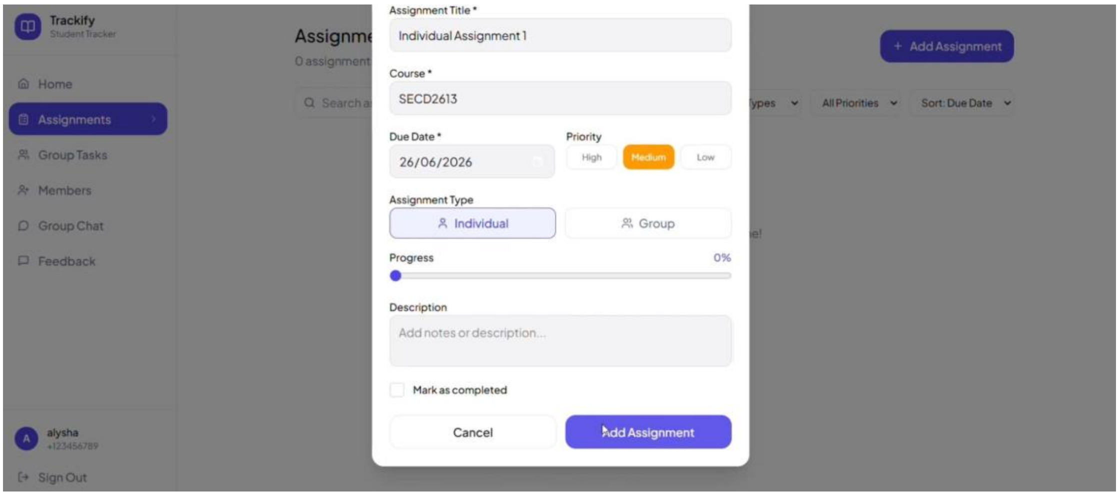


Figure 2 : Add assignment requirement

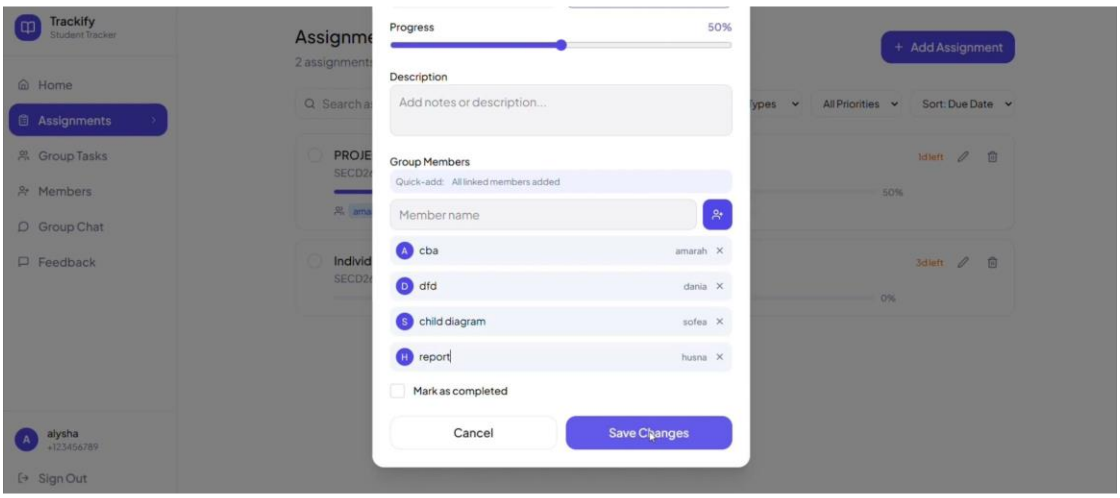


Figure 3 : Add to-do list

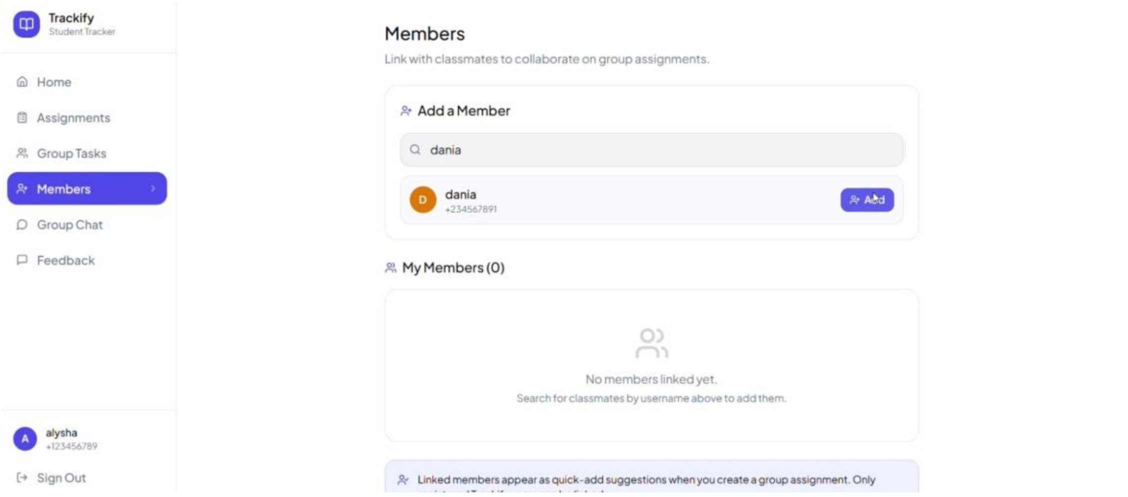


Figure 4 : Add member into group chat

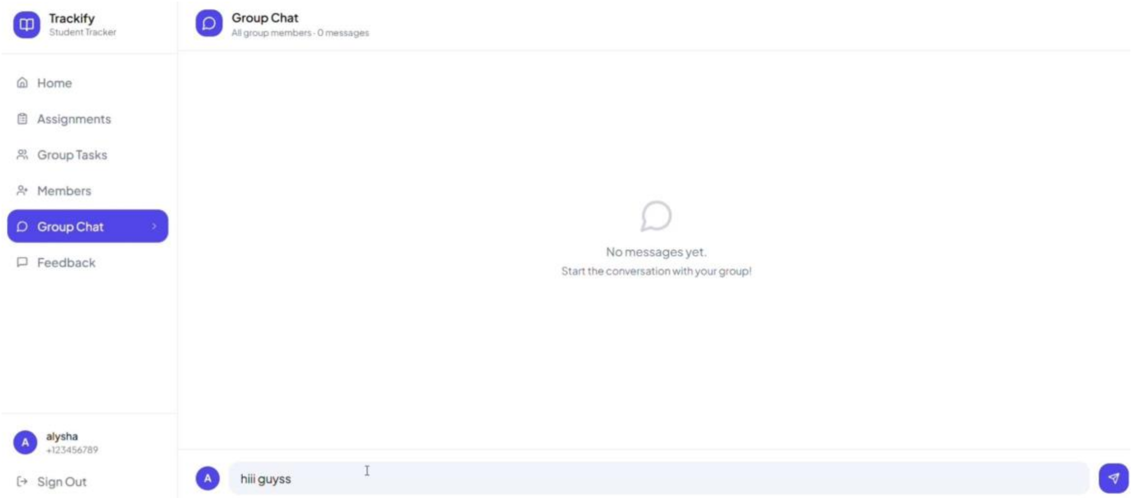


Figure 5 : Send message in the group chat

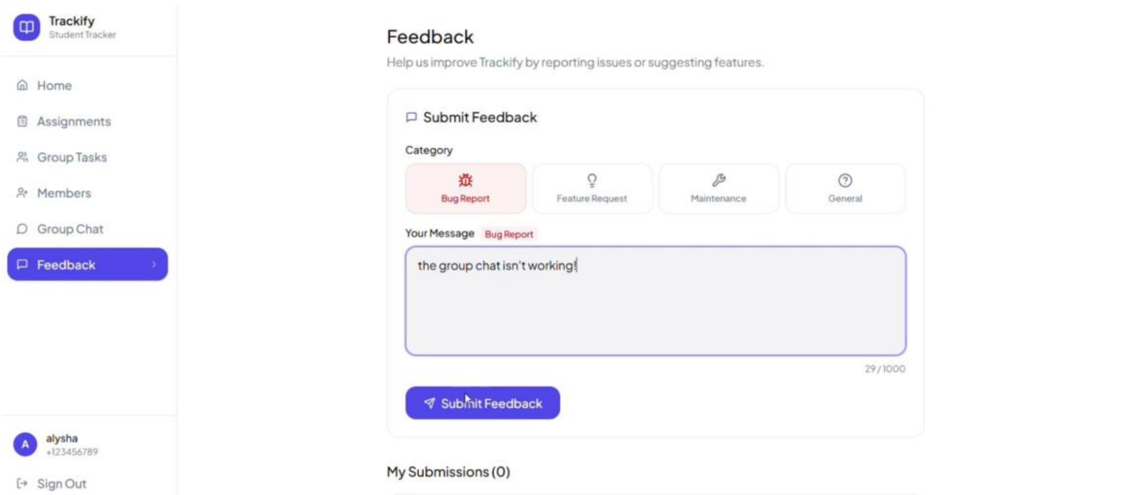


Figure 6 : Send feedback about Trackify

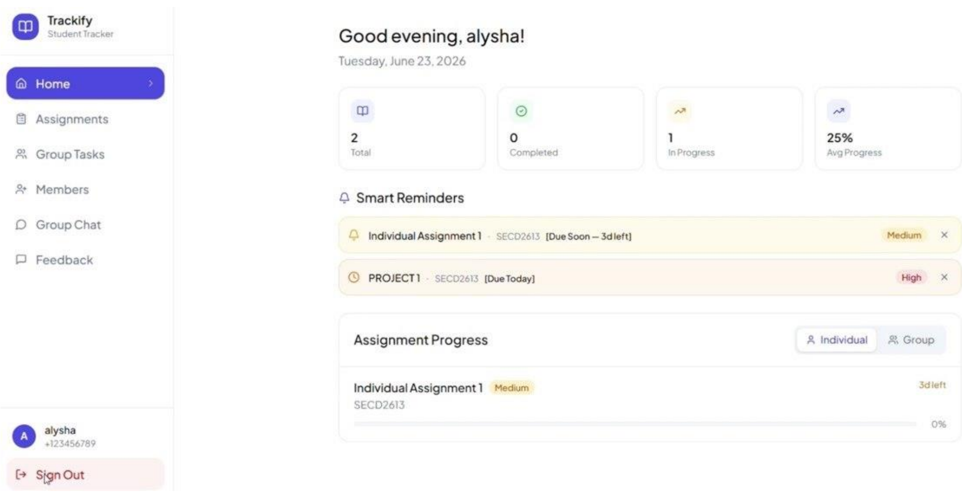


Figure 7 : Trackify dashboard

8.2 Output Design

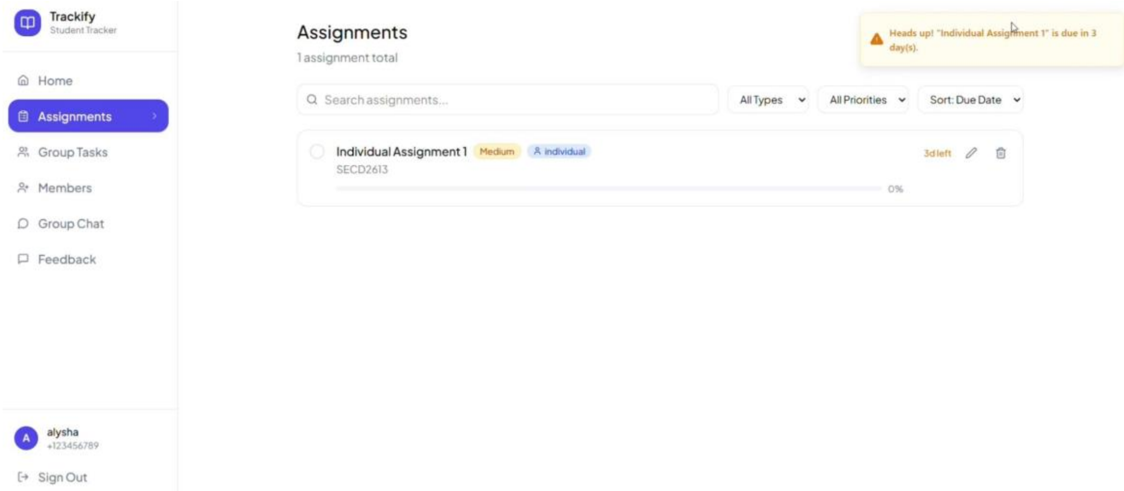


Figure 8 : Notification reminder

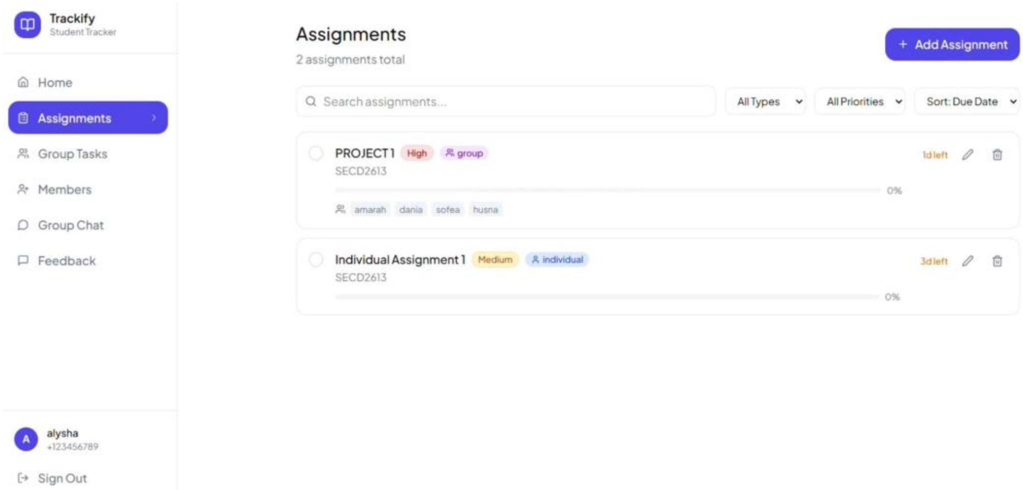


Figure 9 : Assignment list based on due date priority

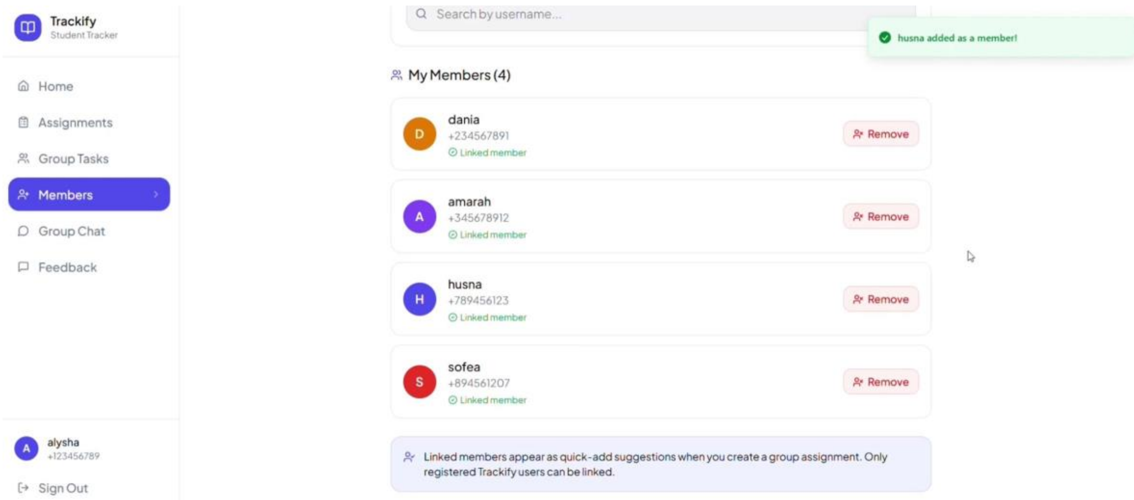


Figure 10 : Group member details

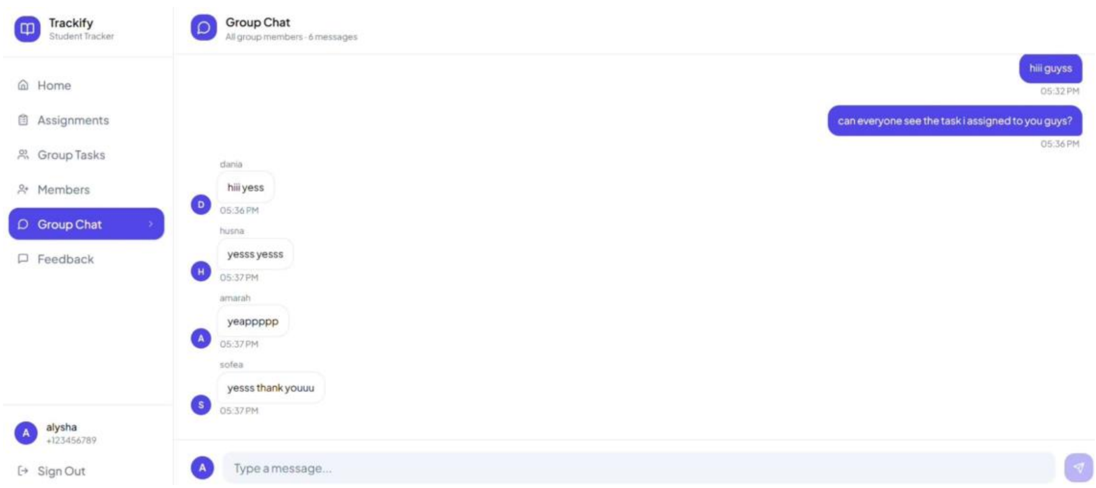


Figure 11 : Group chat Messaging

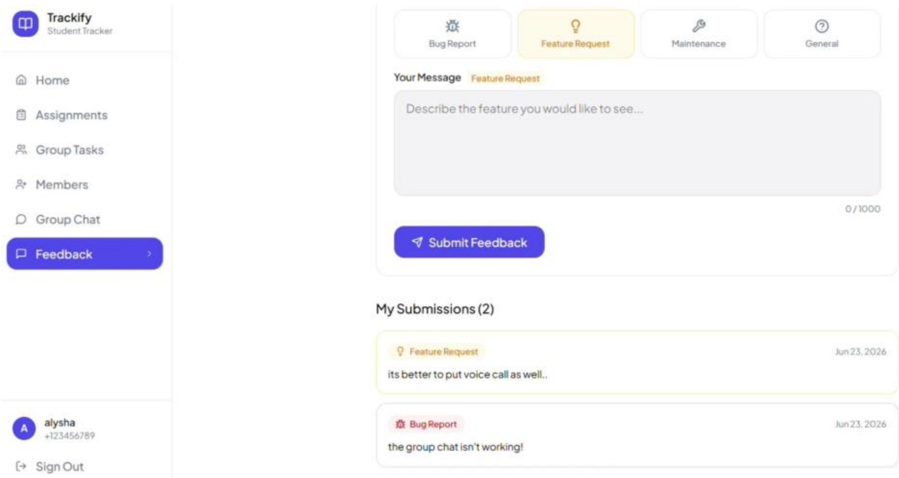


Figure 12 : Feedback Submission

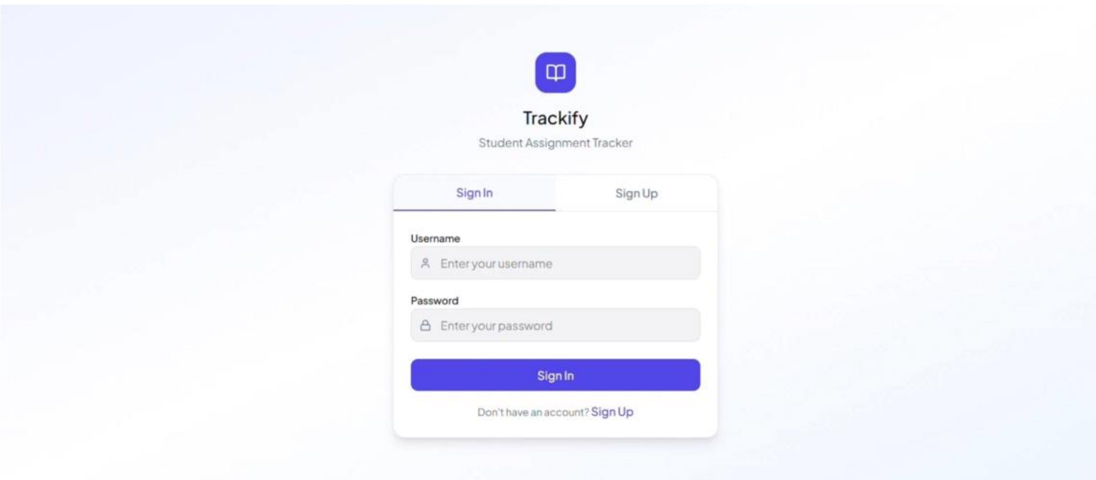


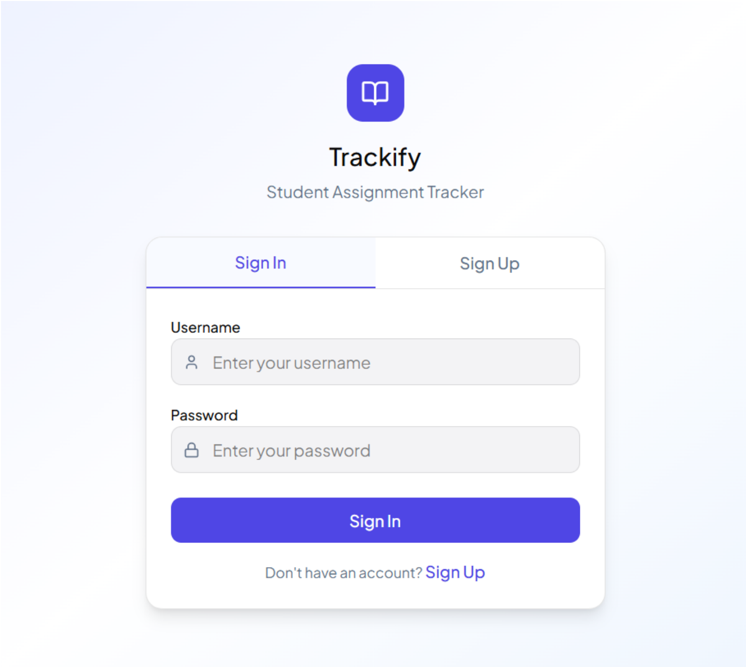
Figure 13 : Login user account

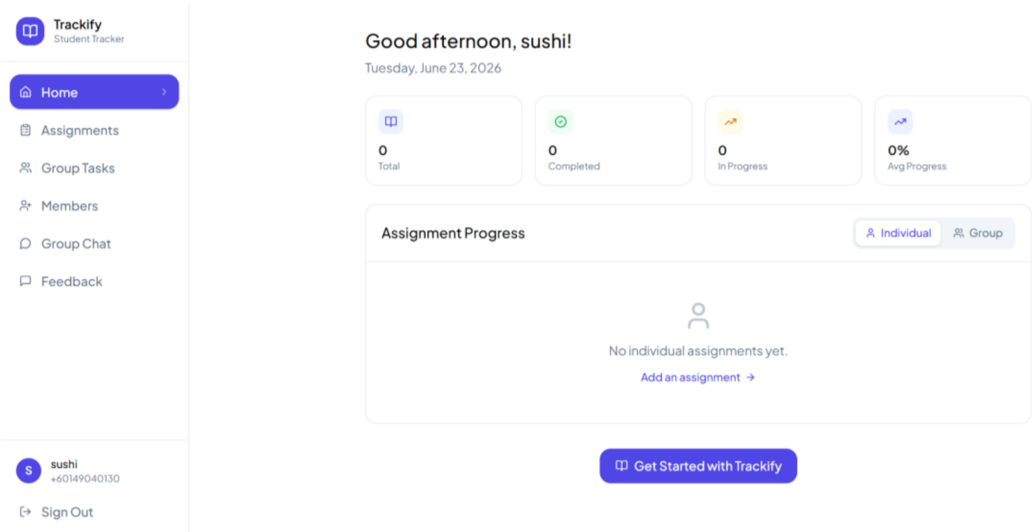
8.3 Prototype

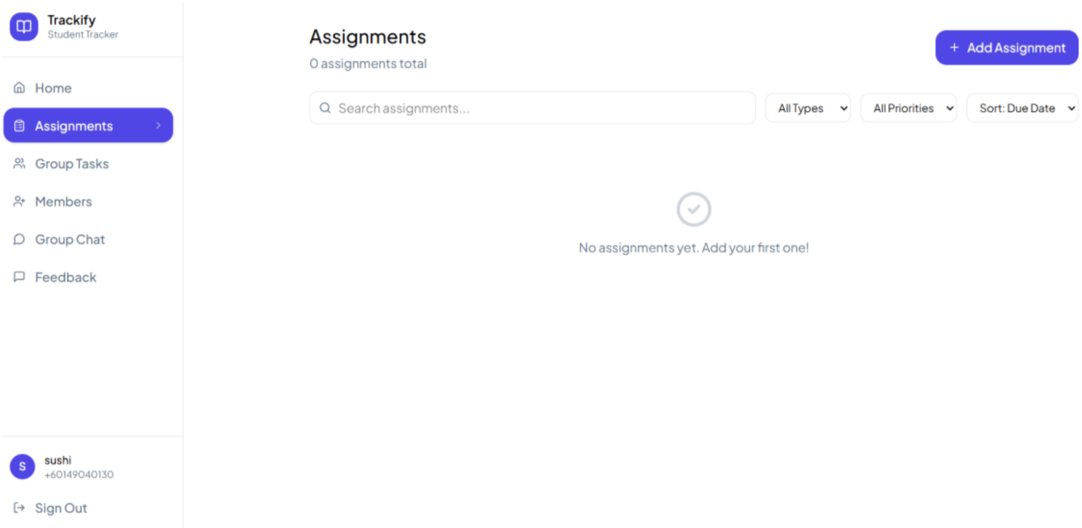
DFD Presentation Link: https://youtu.be/IEb2PmNhCdE

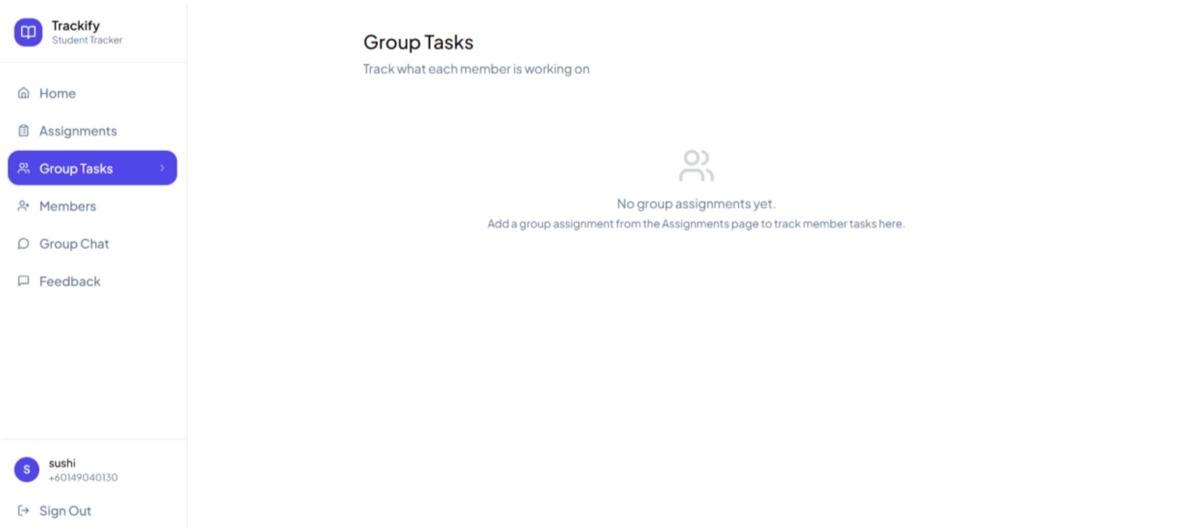
Prototype Presentation link: https://youtu.be/0OVRpr40wgM

Trackify Web : Assignment Tracker Home Screen – Figma Make






Trackify
Student Tracker
Home
Assignments
Group Tasks
Members
Group Chat
Feedback
sushi
+60149040130
Sign Out
Assignments
0 assignments total
+ Add Assignment
Search assignments...
All Types
All Priorities
Sort: Due Date
No assignments yet. Add your first one!


Trackify
Student Tracker
Home
Assignments
Group Tasks
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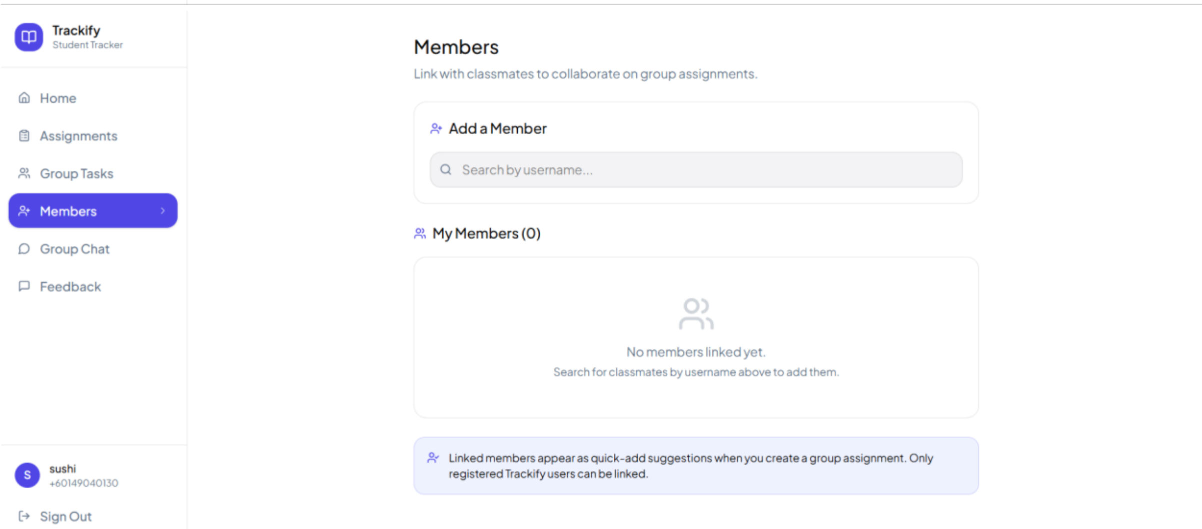
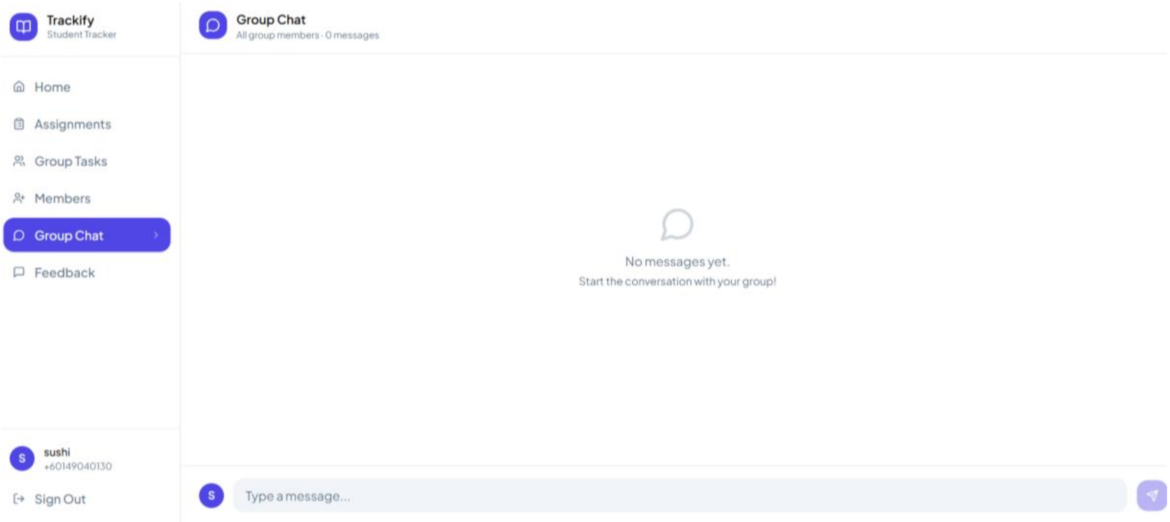
Group Chat
Feedback
sushi
+60149040130
Sign Out
Group Tasks
Track what each member is working on
No group assignments yet.
Add a group assignment from the Assignments page to track member tasks here.


Trackify
Student Tracker
Home
Assignments
Group Tasks
Members
Group Chat
Feedback
sushi
+60149040130
Sign Out
Members
Link with classmates to collaborate on group assignments.
Add a Member
Search by username...
My Members (0)
No members linked yet.
Search for classmates by username above to add them.
Linked members appear as quick-add suggestions when you create a group assignment. Only registered Trackify users can be linked.


Trackify
Student Tracker
Home
Assignments
Group Tasks
Members
Group Chat
Feedback
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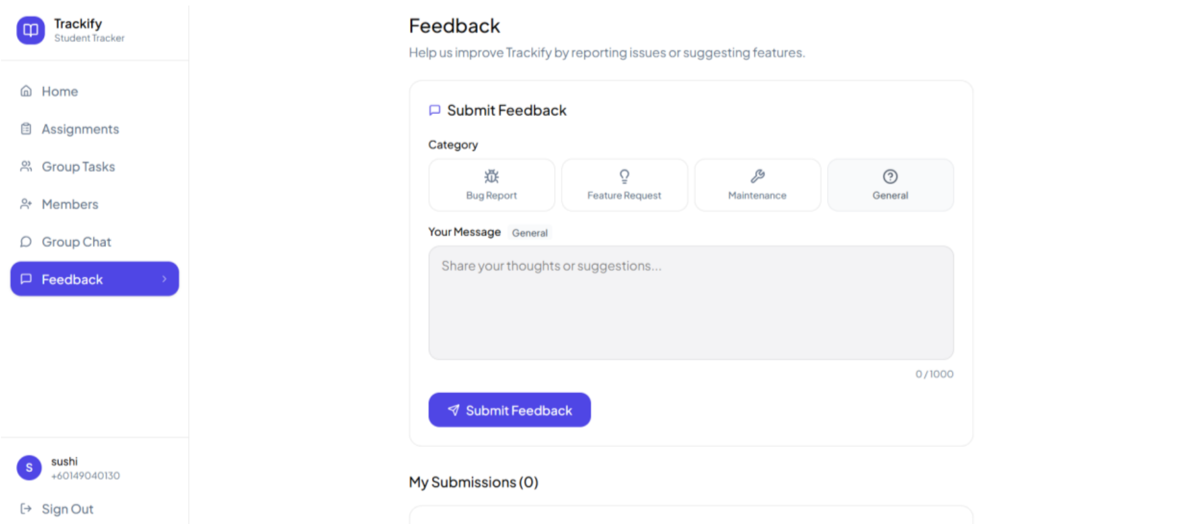
Sign Out
Group Chat
All group members · 0 messages
No messages yet.
Start the conversation with your group!
S
Type a message...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Sign Out
Feedback
Help us improve Trackify by reporting issues or suggesting features.
Submit Feedback
Category
Bug Report
Feature Request
Maintenance
General
Your Message
General
Share your thoughts or suggestions...
0 / 1000
Submit Feedback
My Submissions (0)

9.0 Summary of the proposed system

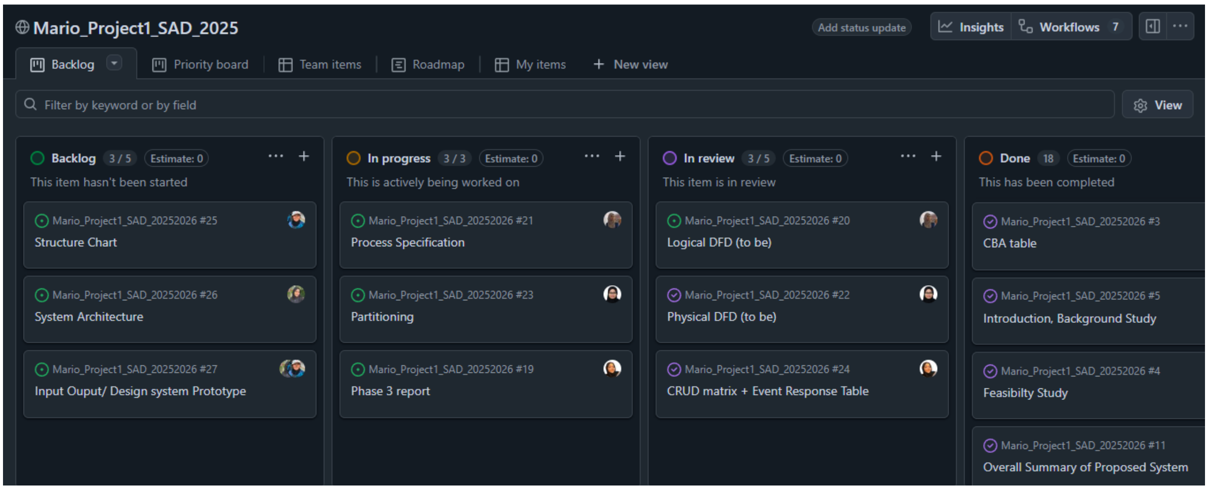
Trackify is designed as a centralized platform to help students manage assignments and group projects more effectively. By integrating smart reminders, progress tracking, group collaboration tools, and a priority ranking system, the platform addresses common issues such as missed deadlines, poor communication, and scattered information across multiple apps. The objective is to reduce stress, improve time management, and enhance academic performance by providing students with clear, organized view of their responsibilities.

The requirement analysis highlights both functional and non-functional needs, ensuring the system is practical and reliable. Students will be able to register, create and update assignments, track progress visually, and collaborate with peers in real time. At the same time, the system emphasizes usability, security, and accessibility, with features like fast loading times, encrypted authentication, and support for future updates. Overall, Trackify is developed to enhance efficiency, accountability, and seamless academic collaboration.

10.0 GitHub Information

URL:

- https://github.com/sofeasof/Mario_Project1_SAD_20252026 (Repository)
- https://github.com/users/sofeasof/projects/1 (Project)



Snapshot of the repository/project during phase 3.